Main Title

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COURSE

Coursenumber

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Contents

1 Introduction **2**
- 1.1 Expectations . . . 2
- 1.2 Vehicle Specifications . . . 2
- 1.3 Powertrain . . . 3

2 Design of the Powertrain **4**
- 2.1 Energy Source . . . 4
- 2.2 Power Conversion . . . 5
- 2.3 Motor and Transmission . . . 5
- 2.4 Auxilary Power . . . 5

3 Motor, Engine, Transmission **6**

4 Energy Source **7**

5 Power Conversion **8**

6 Charging Islands **9**
- 6.1 Rationale for 25 kV AC Overhead Catenary Power Transfer . . . 9
- 6.2 Solar PV Generation and Environmental Adaptation . . . 10

7 Future Developements **12**

8 Simulation **13**

9 Recap **14**

References **15**

1 Introduction

1.1 Expectations

At the beginning of this project, we explored several concepts for an electric vehicle, aiming to go beyond the standard passenger car. We initially considered a performance-oriented Formula E platform as well as a high-efficiency commuting car. However, since individual commuting is fundamentally inefficient as a concept, we looked for more impactful solutions. Rail transport, for instance, is significantly more efficient per ton-kilometer. This led us to combine our interests in performance and efficiency by investigating the electrification of heavy-haul freight trains.
Our focus shifted to the famous Mauritanian Iron Ore Train, which transports ore from inland mines to the coast. This project is particularly compelling because of the theoretical abundance of solar energy in the region. However, the Sahara's harsh desert environment presents a major engineering challenge for electrification. Currently, the train relies on diesel-electric locomotives operating in a series-hybrid configuration [1]. In 2023, the emissions from this operation exceeded 75,000 tons of CO2 equivalents (CO2e), which is roughly equal to the annual carbon footprint of 7,500 average EU households [2, 3].
Our research also showed that we are not the first to explore this potential. During the 2023 United Nations Climate Change Conference (COP28), a memorandum was signed to explore electrifying the rail corridor using green hydrogen [2] which lead us to the conclusion that there is a real use case for this project.

1.2 Vehicle Specifications

Table 1.1: Vehicle specifications of the iron ore train [1].

Purpose	Transport of Iron Ore
Length	2500 m
Weight	5500 t (empty) - 22000 t (full)
Top Speed	65 km/h (empty) - 45 km/h (full)
Max. Gradient	0.5%
Acceleration	0.1 m/s^2
Track Length	700 km

The train is operated in several configurations, typically consisting of four diesel locomotives pulling between 200 and 220 ore wagons, along with two additional carriages for mining staff. The power requirements for the operation can be divided into two distinct segments. On the outbound journey to the mines, the train is empty but must overcome a steady positive gradient, as the mines sit at an elevation of approximately 390 m above sea level [4].
On the return journey to the port, the situation is reversed. Although the train is significantly heavier due to the ore load, much of the route follows a negative gradient, which reduces the net energy required for propulsion. However, the "main crux" of the entire track is a relatively steep uphill section located shortly after leaving the mines toward the port Figure 1.1. This specific segment is a critical design parameter, as it determines the peak power the locomotives must provide to move the fully loaded mass. Using elevation data from the various train stops, a gradient map of the track was

extracted; this profile will serve as the basis for determining the train's specific drive cycle in the following sections.

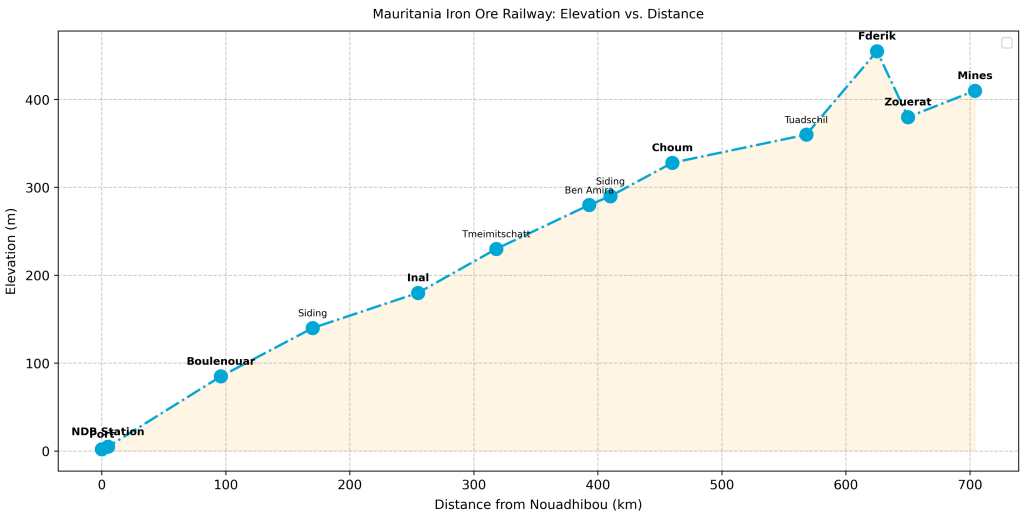

Figure 1.1: The gradient map of the track, extracted from the elevation of the stops along the route.

1.3 Powertrain

Since the train already operates as a series hybrid, we intend to utilize this existing architecture to our advantage. In the current configuration, a diesel generator provides electricity to the traction motors that drive the axles. Our approach is to remove the diesel generator and replace it with a large-scale energy storage system, effectively converting the locomotive into a Battery Electric Vehicle (BEV).

2 Design of the Powertrain

The design of the BEV Mauritanian Iron Core Train will consist of three locomotives, carrying the batteries, motors, and power conversion hardware. In this section, the design of the electric powertrain will be discussed: from the battery and the motor controllers to the motors and the transmission, including the auxiliary power used onboard. This will be done based on design trade-offs, calculations, and simulated results.

2.1 Energy Source

FLXdrive is the perfect example of a full battery-electric locomotive, carrying a total capacity of up to 7 MWh, showing that our ideas are feasible [5].
Starting on a cell level, different chemistries were considered. The main objectives are:

- The cells should have a long lifecycle, as this train will drive the whole day, every day. Any downtime directly results in a loss of income for every party involved.
- The cells should be able to operate in rough temperature conditions. During the day, temperatures can rise to a stunning 50℃, whereas they might fall below 0℃ during a winter night.

LCO batteries have a high power density and a somewhat short lifetime. Since the design is not optimizing for weight or volume, LCO batteries were discarded. The comparison was ultimately made between LFP and LTO battery cells. LFP is broadly used for EVs and energy storage applications, even though its characteristic flat SOC curve makes it difficult to track the battery level. LTO is known for its extreme temperature endurance, fast charging capabilities, and unmatched lifespan. Even though LFP is cheaper to purchase, the longer lifecycle of LTO will make that chemistry cheaper in the long term than LFP. The longer lifetime (30,000 cycles), together with the better temperature performance, made us choose the Lithium Titanate Oxide Battery Cell - LTO 2.3V 40Ah cylindrical cell [6]. The cylindrical cell was chosen, rather than the pouch cell, as it is a safer option in terms of robustness, and the weight reduction from the pouch cell is not part of our requirements.
The total onboard energy required is 12 MWh (see section ...). This will be equally divided over the three locomotives, with each carrying a battery of 4 MWh. To reach the 800 V DC bus voltage, 348 cells will be placed in series. A total of 125 parallel connections will be made, resulting in a capacity of 5000 Ah. A single cell is estimated at the price of 39 euros [7]. The total amount of money for all 43,500 cells accumulates to 5 million euros.
The total cell weight will be 54,375 kg per battery, with a volume of 30 m^3 (including terminals). At the cell level, the 4MWh capacity per locomotive yields an energy-to-weight ratio of approximately 73.5Wh/kg. Given the LTO cell's 10C discharge capability, the theoretical maximum power output is 40MW per locomotive, resulting in a power-to-weight ratio of 736W/kg. There is also enough space for the battery in the locomotive. Regarding the placement, the battery packs will be housed centrally, occupying the space traditionally reserved for the diesel engine. The modules will be securely mounted in shock-resistant racks.
The LTO cell can handle charge and discharge rates of 10C, according to its datasheet [6]. The highest energy requirement to fully charge the battery occurs at stop 5 (see section ...). The energy required is almost 10 MWh over a timespan of 30 minutes, so the peak charge power will be 20 MW. Therefore, the current will reach a maximum of 12,800 A (taking the discharge cut-off voltage of 1.5 V into account), or 2.56C, staying well below the current limit. Regenerative braking also won't deliver

higher charge rates, since the downhill gradient is very shallow and the braking distance is very long. As this application integrates a BEV, the tank-to-wheel emissions are zero. Furthermore, since the electricity generation is completely decoupled from any grid and 100% of the energy is generated by solar power, the well-to-tank emissions are also zero. Thus, the expected well-to-wheel emissions are zero.

2.2 Power Conversion

2.3 Motor and Transmission

2.4 Auxilary Power

3 Motor, Engine, Transmission

4 Energy Source

5 Power Conversion

6 Charging Islands

6.1 Rationale for 25 kV AC Overhead Catenary Power Transfer

For the dynamic charging islands, the system must reliably transfer between 10 MW and 20 MW of power to a moving locomotive in a 50°C Sahara Desert environment. After evaluating both inductive charging and overhead catenary systems, the overhead catenary architecture was selected as the only physically and mathematically feasible option. The selection is driven by critical thermal, electromagnetic, and electrical current constraints.

Thermal Management and Heat Dissipation

Assuming a highly optimistic 95% efficiency for a 20 MW power transfer, the system must continuously dissipate approximately 1 MW of waste heat.

- **Track-Side Infrastructure:** Inductive transmitter coils must be buried in desert sand or concrete, which act as thermal insulators. This creates a "thermal trap" that would require unfeasible underground liquid cooling to prevent the coils from melting their own insulation. Conversely, an overhead catenary system distributes this heat across 5 km of bare copper wire suspended in the open air, resulting in a highly manageable thermal load of approximately 200 W/m that is safely dissipated via passive ambient air cooling.
- **Power Transfer Interface:** Inductive receiver pads possess a fixed, enclosed surface area that would act as a concentrated heat sink, requiring massive onboard liquid cooling. The overhead pantograph mitigates contact-point melting through *kinetic cooling*; by manually constraining the train's charging speed to a minimum of 25 km/h, the pantograph continuously slides across "fresh," unheated wire, spreading the friction and electrical arcing heat dynamically along the track length.

Electromagnetic Safety and Chassis Integrity

Transferring up to 20 MW of power wirelessly via induction generates massive stray magnetic fields. At this power scale, the magnetic flux would induce significant eddy currents in the train's steel chassis, the iron wheels, and the rails, effectively heating the locomotive structure like an induction stove and creating severe electrocution hazards. The overhead catenary system isolates the electrical current within dedicated, shielded high-voltage cables leading directly to the onboard rectifiers, keeping the thermal load strictly contained within the train's dedicated active liquid-cooling loops.

Transmission Standard and Current Constraints

While a native low-voltage DC microgrid is highly efficient, the physical limits of the pantograph-to-wire contact point prohibit a purely DC catenary at this power scale.

- **The Amperage Limit:** Standard DC railway voltages (e.g., 750 V DC to 3 kV DC) are practically limited to around 3 MW. Attempting to push 20 MW at 3 kV DC would result in over 6,000 Amps of current. This extreme amperage exceeds the physical limits of standard heavy-rail pantographs and would instantly melt the copper contact point.

- **25 kV AC Standard:** By specifying a 25 kV AC overhead line, the track can comfortably deliver 20 MW while keeping the current strictly under 800 Amps. This falls well within the safe operational limits of standard high-speed and heavy-freight rail pantographs.

The 25 kV AC overhead catenary is the selected architecture because it naturally manages the 1 MW thermal penalty through spatial distribution and passive cooling, avoids electromagnetic interference with the train chassis, and maintains a safe contact current. Furthermore, adopting 25 kV AC aligns the microgrid with proven global industry standards for heavy freight and high-speed rail, ensuring commercial component availability, interoperability, and long-term mechanical reliability in the harsh Sahara environment.

6.2 Solar PV Generation and Environmental Adaptation

The energy generation subsystem is the primary power source for the off-grid charging islands. Due to the extreme environmental conditions of the Sahara Desert and the mechanical stresses of heavy-haul rail operations, a highly specialized PV architecture is required to ensure long-term system reliability.

1. Mechanical and Environmental Placement Strategy

The PV arrays are intentionally set back 20–30 meters from the track-side rather than being placed between or directly adjacent to the rails. This design choice is necessitated by two primary factors:

- **Vibration Mitigation:** The 22,000-tonne iron-ore trains generate significant ground-borne vibrations. Direct rail-side placement would lead to accelerated cell micro-cracking and solder-joint failure. A 20m setback provides sufficient soil-damping to protect the module's crystalline structure.
- **Dust and Occlusion Management:** Moving trains continuously shed iron-ore dust and lubricant aerosols. These particles are highly opaque and electrically conductive; buildup on modules would lead to severe shading losses and potential fire hazards. The setback minimizes the volume of train-generated particulates reaching the array surface.

2. Module Technology: Heterojunction (HJT) and Bifaciality

Standard PERC (Passivated Emitter and Rear Cell) modules are rejected for this project due to their high temperature coefficients. Instead, the design specifies **Bifacial HJT Modules**.

- **Thermal Resilience:** HJT cells possess a superior temperature coefficient of approximately -0.25%/°C. In desert conditions where module temperatures frequently exceed 70°C, HJT panels maintain significantly higher conversion efficiency compared to standard technologies that lose $\sim$15% of their rated power in similar heat.
- **Albedo Utilization:** The Sahara sand provides a high albedo (reflectivity) of 30–45%. Bifacial modules capture reflected irradiance on their rear surface, providing a "free" power boost of 15–20% without increasing the physical footprint of the array.

3. Maintenance and Autonomous O&M

In the desert, sand accumulation (soiling) is the single largest cause of yield degradation, potentially reducing output by up to 30% if left uncleaned. To ensure consistent 10 MW delivery:

- **Robotic Cleaning:** The arrays are equipped with autonomous, waterless robotic cleaners. These robots perform nightly dry-sweeping cycles to remove sand and dust without consuming local water resources or causing abrasive damage to the anti-reflective coatings.

4. Sizing and Auxiliary Load Management

To ensure the system is self-sustaining, the total peak capacity (kW_p) of the array is oversized by 10–15% beyond the charging requirements of the train. This overhead is dedicated to powering the **Auxiliary Power Supply (APS)**, which drives the active liquid-cooling pumps and energy management systems required to keep the stationary LTO battery banks within their safe operational temperature window during daylight hours.

Table 6.1: Solar PV Engineering Design Summary

Design Element	Selection	Engineering Rationale
Placement	20–30m Setback	Vibration damping and iron-ore dust mitigation.
Cell Type	Heterojunction (HJT)	Low temp. coefficient (-0.25%/°C) for 50°C ambient.
Module Style	Bifacial	Utilizes 30–45% sand albedo for 15–20% yield boost.
Cleaning	Waterless Robotics	Prevents up to 30% yield loss from desert soiling.
Sizing	10–15% Oversize	Dedicated capacity for battery liquid-cooling loads.

7 Future Developements

8 Simulation

9 Recap

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